

3 Basic methods for computing probabilities

In order to proceed we must be equipped with some basic methods for computing probabilities. This is the aim of this section.

3.1 Law of total probability

In the following we let a probability space $(\Omega, \mathbb{P})$ be given. A (finite or infinite) sequence $(B_i)_{i \geq 1}$ of events is said to be a *partition* of Ω if the events are disjoint (i.e., $B_i \cap B_j = \emptyset$ for $i \neq j$) and $\Omega = \bigcup_{i \geq 1} B_i$. Intuitively, a partition provides a (possibly infinite) number of cases such that exactly one holds for each outcome. The *law of total probability*, stated below, finds the probability of a given event by *conditioning* with respect to the partition. It boils down to a sequence of “if-then” questions and evaluating the resulting conditional probabilities. It captures the main idea of conditioning which is very useful throughout probability theory.

Proposition 3.1 (Law of total probability). *Let $(B_i)_{i \geq 1}$ be a partition of Ω . Then for any event A we have*

$$\mathbb{P}(A) = \sum_{i \geq 1} \mathbb{P}(A|B_i)\mathbb{P}(B_i). \quad (3.1)$$

Proof. Write

$$A = A \cap \Omega = A \cap \left(\bigcup_{i \geq 1} B_i \right) = \bigcup_{i \geq 1} (A \cap B_i),$$

and note that the last union is disjoint. By the sum rule, we have

$$\mathbb{P}(A) = \sum_{i \geq 1} \mathbb{P}(A \cap B_i).$$

By the product rule, we have, for each i ,

$$\mathbb{P}(A \cap B_i) = \mathbb{P}(A|B_i)\mathbb{P}(B_i).$$

(Recall our convention that the product rule holds even if $\mathbb{P}(B_i) = 0$.) This completes the proof of (3.1). $\square$

The following is an elementary example of the law of total probability. It also illustrate some rather subtle points.

Example 3.2. There are two fair coins, say A and B . Coin A is a normal coin (head and tail) but both sides of coin B are heads. A coin is chosen at random and then flipped. Find the probability that we observe heads.

Solution. Consider the events

$$E_1 = \{\text{coin } A \text{ is chosen}\}, \quad E_2 = \{\text{coin } B \text{ is chosen}\}, \quad E_3 = \{\text{observe heads}\},$$

which are subsets of some unspecified sample space Ω . The problem statement amounts to the following assumptions:

$$E_1 \cup E_2 = \Omega, \quad E_1 \cap E_2 = \emptyset, \quad (3.2)$$

$$\mathbb{P}(E_1) = \mathbb{P}(E_2) = \frac{1}{2}, \quad \mathbb{P}(E_3|E_1) = \frac{1}{2}, \quad \mathbb{P}(E_3|E_2) = 1. \quad (3.3)$$

By assumption, E_1 and E_2 form a partition of Ω . By the law of total probability, we have

$$\mathbb{P}(E_3) = \mathbb{P}(E_3|E_1)\mathbb{P}(E_1) + \mathbb{P}(E_3|E_2)\mathbb{P}(E_2) = \frac{1}{2} \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{3}{4}.$$

Having solved the problem, we discuss the hidden subtleties. First, our computation applies to *any* probability space $(\Omega, \mathbb{P})$ and events E_1, E_2, E_3 which satisfy (3.2) and (3.3). This illustrates the fact that in probability theory we are often more interested in the probabilities (or more precisely the distributions of random variables) rather than how they are realized on specific probability spaces.

Second, in many situations the problem does not specify completely the probability space but only gives certain restrictions such as (3.2) and (3.3). It is important that these restrictions are *consistent*, meaning that they can be satisfied by some probability space. Otherwise, the problem does not make sense and we will obtain contradictions (e.g. we find disjoint events E_4 and E_5 whose probabilities add up to more than 1). Also, the restrictions should be sufficient to determine uniquely the desired quantities.

For this problem, we may construct a probability space as follows:

$$\Omega = \{a_1, a_2, b_1, b_2\}, \quad \mathbb{P}\{\omega\} = \frac{1}{4} \text{ for all } \omega.$$

Here a_1, a_2 correspond to the two sides of coin A (similarly for coin B). Let

$$E_1 = \{a_1, a_2\}, \quad E_2 = \{b_1, b_2\}, \quad E_3 = \{a_1, b_1, b_2\}.$$

It is immediate to check that the conditions (3.2) and (3.3) hold.

3.2 Bayes's rule

Bayes's rule is a simple consequence of the product rule. Nevertheless, it is one of the most important results in probability theory because it determines how probabilities should be updated in light of new information.

Theorem 3.3 (Bayes's rule). *If $\mathbb{P}(B) > 0$ then for any event A we have*

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)}. \quad (3.4)$$

Proof. By the product rule, we have

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

□

Remark 3.4 (Baye’s rule and plausible reasoning). In Section 1.1.2 we mentioned that probability theory may be regarded as an extension of logic. To give a flavor of some ideas involved, consider the following induction:

- Suppose A implies B and we observe that B is true. Then, A becomes more probable.

Let us justify this argument using Bayes’s rule. Before we learn whether B is true, the plausibility of A is $\mathbb{P}(A)$. After we know that B is true, the plausibility becomes $\mathbb{P}(A|B)$. From Bayes’s rule (3.4), $\mathbb{P}(A|B)$ is the multiple of $\mathbb{P}(A)$ and the *likelihood ratio* $\frac{\mathbb{P}(B|A)}{\mathbb{P}(B)}$. Since A implies B by assumption, we have $\mathbb{P}(B|A) = 1$. It follows that $\frac{\mathbb{P}(B|A)}{\mathbb{P}(B)} \geq 1$ and we have

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)}{\mathbb{P}(B)}\mathbb{P}(A) \geq \mathbb{P}(A).$$

That is, A becomes more probable after observing that B holds, and the extra amount (which may be zero) is quantified by Bayes’s rule. $\square$

Example 3.5. Consider the situation of Example 3.2. Suppose we are *only* told that the result is “heads”. For us, what is the conditional probability that coin A was chosen and tossed?

Solution. Using the notations in Example 3.2, the quantity of interest is $\mathbb{P}(E_1|E_3)$. By Bayes’s rule, we have

$$\mathbb{P}(E_1|E_3) = \frac{\mathbb{P}(E_3|E_1)}{\mathbb{P}(E_3)}\mathbb{P}(E_1) = \frac{1}{3/4} \frac{1}{2} = \frac{2}{3}.$$

Given the observation, the “posterior probability” $\frac{2}{3}$ is larger than the “prior probability” $\frac{1}{2}$. Now read again Remark 3.4.

*Example 3.6.*¹ A test for whether someone has COVID is 90% sensitive and 80% specific, meaning it leads to 90% true “positive” results for COVID patients and 80% true negative results for uninfected people (but also generates 20% false positives for uninfected people). Assuming 5% of the population actually has COVID, what is the probability that a randomly chosen person who tests positive is really a COVID patient?

Solution. The information given in the problem allows us to write down a set of probabilities. Consider the events

$$U = \{\text{the person has COVID}\}, \quad P = \{\text{the test gives a positive result}\}.$$

Then

$$\mathbb{P}(P|U) = 0.9, \quad \mathbb{P}(P|U^c) = 0.2, \quad \mathbb{P}(U) = 0.05.$$

We also compute, using the law of total probability,

$$\mathbb{P}(P) = \mathbb{P}(P|U)\mathbb{P}(U) + \mathbb{P}(P|U^c)\mathbb{P}(U^c) = 0.9 \cdot 0.05 + 0.2 \cdot 0.95 = 0.235,$$

¹The numbers are taken from https://en.wikipedia.org/wiki/Bayes_theorem and do not represent the accuracy of actual COVID tests.

a probability which is required to apply Bayes's rule.

We are interested in the conditional probability $\mathbb{P}(U|P)$. By Bayes's rule, we have

$$\mathbb{P}(U|P) = \frac{\mathbb{P}(P|U)}{\mathbb{P}(P)}\mathbb{P}(U) = \frac{0.9}{0.235}0.05 \approx 0.19.$$

Thus, even if someone tests positive, the (conditional) probability that the subject actually has COVID is only 19% (which is higher than 5%). This is because in this group only 5% of people are infected, and most positives are false positives coming from the remaining 95% of healthy people.

3.3 Elements of combinatorics

Combinatorics is the study of *counting*. Here are two related ways in which counting arises in probability theory:

- (i) Consider a probability space $(\Omega, \mathbb{P})$ where Ω is *finite* and $\mathbb{P}$ is *uniform* on Ω . That is, for any $\omega \in \Omega$ we have $\mathbb{P}\{\omega\} = \frac{1}{|\Omega|}$ where $|\Omega| < \infty$ is the number of elements of Ω . Then for any event $A \subset \Omega$ we have

$$\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}\{\omega\} = \sum_{\omega \in A} \frac{1}{|\Omega|} = \frac{|A|}{|\Omega|}.$$

So evaluating $\mathbb{P}(A)$ is equivalent to counting the number of elements of A (and Ω). Table 1.1 shows a simple example where $|\Omega| = 8$.

- (ii) More generally, an event A may be expressed, possibly by a symmetry argument, as a finite disjoint union $A = \bigcup_{i=1}^k A_i$ such that all A_i have the *same* probability, say p . Then $\mathbb{P}(A) = \sum_{i=1}^k \mathbb{P}(A_i) = kp$. Typically some counting is needed to find k . This technique arises, for example, in the derivation of the *binomial distribution* (see Example 3.11 and Theorem 4.36).

There are two basic counting rules. In what follows the sets involved are assumed to be finite.

- (Sum rule) If $A = \bigcup_{i=1}^k A_i$ is a disjoint union, then

$$|A| = \sum_{i=1}^k |A_i|.$$

- (Product rule) If the elements of A are in one-to-one correspondence with a product set $A_1 \times \cdots \times A_k$, then

$$|A| = \prod_{i=1}^k |A_i|.$$

The above sum and product rules of counting are related to, but must be distinguished from, the sum and product rules of probability.

Example 3.7. Consider a 3-course dinner. For the appetizer we can choose from {salad, soup}. For the main course we have one of {steak, fish, tofu}. Finally, we choose a desert from {icecream, cheese cake, apple pie}. What is the number of possible choices?

Solution. Each choice is uniquely associated to a triple of the form

$$(\text{appetizer}, \text{main course}, \text{desert}).$$

By the product rule, the number of possible choices is $2 \times 3 \times 3 = 18$. It is important to note that for the product rule to hold, the number of choices available in each step must *not* depend on the choices made in the previous steps. Suppose we may also choose lobster tail (which is more expensive) as the main course but if we choose lobster we cannot have a desert. Then the product rule cannot be applied directly.

Many counting problems involve *choosing elements from a given set*. To be more specific, consider a set $A = \{a_1, \dots, a_n\}$ containing n *distinct* elements $a_1, \dots, a_n$. Suppose $k \leq n$ distinct elements of A are chosen. There are two ways to describe the result:

- Give an *ordered k -tuple* $(b_1, \dots, b_k)$ where b_1 is the first element chosen, b_2 is the second element chosen, and so on. When $k = n$, we say that $(b_1, \dots, b_n)$ is a *permutation* of the elements of A .
- Give an *unordered set* $\{b_1, \dots, b_k\}$ containing k elements of A . In other words, $\{b_1, \dots, b_k\}$ is a subset of A containing k elements.

Example 3.8. Let $A = \{1, 2, 3, 4, 5\}$ and $k = 3$. A possible *ordered* result is $(2, 1, 4)$. The *unordered* set $\{1, 2, 4\}$ correspond to 6 ordered results:

$$\{1, 2, 4\} \leftrightarrow (1, 2, 4), (1, 4, 2), (2, 1, 4), (2, 4, 1), (4, 1, 2), (4, 2, 1) \quad (3.5)$$

In particular, there are 6 permutations of three distinct elements.

The *factorial* $n!$ is defined for a non-negative integer n by $0! = 1$ and

$$n! = n \cdot (n-1) \cdots 1, \quad n \geq 1. \quad (3.6)$$

For example, we have $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$. Note the identity

$$n! = n \cdot (n-1)!, \quad n \geq 1.$$

It holds even when $n = 1$ since $0! = 1$.

The following theorem is fundamental and must be thoroughly understood.

Theorem 3.9 (Permutation and binomial coefficient). *Let $n \in \{1, 2, \dots\}$.*

- (i) *For $1 \leq k \leq n$, the number of ways to pick k ordered distinct elements from a set with n distinct elements is*

$$n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}. \quad (3.7)$$

In particular, the number of permutations of n distinct elements is $n!$.

(ii) For $0 \leq k \leq n$, the number of ways to pick k unordered distinct elements from a set with n distinct elements is

$$\binom{n}{k} = \frac{n!}{(n-k)!k!}, \quad (3.8)$$

which is called the binomial coefficient.

Proof. (i) Let $A = \{a_1, \dots, a_n\}$ be the set under consideration. Consider the set

$$B = \{(b_1, \dots, b_k) : b_1, \dots, b_k \in A \text{ all distinct}\}.$$

Each element $(b_1, \dots, b_k)$ is a way to pick k ordered elements from A .

To find $|B|$ we note the following:

- There are n ways to choose b_1 .
- Given b_1 , there are $n - 1$ ways to choose b_2 .
- ...
- Given $b_1, \dots, b_{k-1}$, there are $n - k + 1$ ways to choose b_k .

By the product rule, $|B|$, the size of B or equivalently the number of ways to choose $(b_1, \dots, b_k)$, is $n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}$. Note that defining $0! = 1$ allows the equality in (3.8) to hold even when $k = n$.

(ii) Again we let $A = \{a_1, \dots, a_n\}$ be the set under consideration. We need to find the number of subsets of A with k elements.

First suppose $k = 0$. There is exactly one subset of A which has zero elements, namely the empty set $\emptyset$. This is consistent with the formula

$$\binom{n}{0} = \frac{n!}{0!n!} = \frac{n!}{1 \cdot n!} = 1.$$

Note that the convention $0! = 1$ is used.

Now suppose $1 \leq k \leq n$. Let N be the number of ways of choosing a subset $\{b_1, \dots, b_k\}$ of A with k elements. For each $\{b_1, \dots, b_k\}$, there are $k!$ permutations by (i). Together, there are $N \cdot k!$ ordered k -tuples. By construction, they are all ordered k -tuples of distinct elements of A . From (i) again, we have

$$N \cdot k! = \frac{n!}{(n-k)!} \Rightarrow N = \frac{n!}{(n-k)!k!} = \binom{n}{k}.$$

□

Note that the binomial coefficient satisfies the *symmetry property*

$$\binom{n}{k} = \binom{n}{n-k}, \quad 0 \leq k \leq n. \quad (3.9)$$

This can be seen either directly from the formula (3.8) or from the combinatorial observation that choosing a subset with k elements is equivalent to choosing its complement which has $n - k$ elements. In particular $\binom{n}{n} = \binom{n}{0} = 1$.

We call $\binom{n}{k}$ the *binomial coefficient* because these quantities show up in the *binomial theorem*: For any $a, b \in \mathbb{R}$ and $n \in \mathbb{N}$, we have

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k. \quad (3.10)$$

For example, since $\binom{5}{1} = \binom{5}{4} = 5$ and $\binom{5}{2} = \binom{5}{3} = 10$, we have

$$(1 + t)^5 = 1 + 5t + 10t^2 + 10t^3 + 5t^4 + t^5.$$

For completeness, we give a proof of the binomial theorem in Section A.3.

We end this section with a remark and an example, both of which will be expanded upon in later sections.

Remark 3.10 (Growth of $n!$). The factorial $n!$ *grows extremely fast* as n increases. For example, we have $15! = 1307674368000$, and $70! \approx 1.2 \cdot 10^{100}$ is already much larger than current estimates of the number of protons in the “observable universe”, which is about 10^{80} .² So, in most cases it is infeasible to solve combinatorial problems by brute force. The exact growth rate of $n!$ is given by *Stirling’s formula* which is briefly discussed in Section A.3.

Example 3.11 (The symmetric binomial distribution). Suppose a fair coin is tossed n times. We use the sample space $\Omega = \{0, 1\}^n$ and let $\mathbb{P}$ be uniform on Ω , i.e., $\mathbb{P}\{\omega\} = \frac{1}{2^n}$ for each $\omega \in \Omega$. Let $X(\omega) = \omega_1 + \cdots + \omega_n$ be, say, the number of heads (this generalizes Table 1.1). For any integer k with $0 \leq k \leq n$, we have

$$\mathbb{P}(X = k) = \frac{|\{X = k\}|}{|\Omega|} = \frac{|\{X = k\}|}{2^n}.$$

Consider the event

$$\{X = k\} = \{\omega = (\omega_1, \dots, \omega_n) \in \{0, 1\}^n : \omega_1 + \cdots + \omega_n = k\}.$$

Any sample point ω in $\{X = k\}$ gives exactly k times at which the toss gives heads, and so corresponds to a subset of $\{1, \dots, n\}$ with k elements, and vice versa. By Theorem 3.9(ii), we have $|\{X = k\}| = \binom{n}{k}$. This gives

$$\mathbb{P}(X = k) = \binom{n}{k} 2^{-n}, \quad k = 0, 1, \dots, n. \quad (3.11)$$

This probability mass function gives a special case of the *binomial distribution* (Definition 4.37).

Note that since $\mathbb{P}(X = 0) + \cdots + \mathbb{P}(X = n) = 1$, summing (3.11) over k gives the identity

$$\binom{n}{0} + \cdots + \binom{n}{n} = 2^n. \quad (3.12)$$

It also follows from the binomial theorem (3.10) by letting $a = b = 1$.

²See https://en.wikipedia.org/wiki/Observable_universe.